σ-Additive Probability on Orthomodular Lattices: A Survey and an Open Problem

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Survey note — June 2026

Abstract

The passage from a finitely coherent assignment of probabilities to a countably additive measure is, in the Boolean case, governed by a single condition (countable additivity) and executed by classical machinery (Carathéodory; Loomis–Sikorski; Stone duality). On a *non-distributive* orthomodular lattice (OML) — the algebra of propositions of a quantum system — this machinery breaks, and the question of when finite coherence forces countable behaviour is open. We survey the problem at the level needed to begin work on it. After fixing the apparatus (orthomodular lattices, the gap between orthogonality and meet-zero, the state/meet-additive/charge ladder, the McDonald–Bimbó dual space), we separate the question into two axes. The *extension* axis is classical (Horn–Tarski feasibility), settled but for a single residual case — singular states on $L(H)$; the *descent* axis is genuinely open and rests on a representation that does not exist: a Loomis–Sikorski theorem for σ-complete OMLs. The stake is more than technical: such a representation would furnish a *point-free*, σ-additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory — built from the entailment relation rather than descended from a sample space. We review what is known and what is ruled out, and close with the two directions in which the open problem might be approached.

1 Introduction

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*: that mass cannot persist along a vanishing sequence of events. De Finetti [5] held that only finite additivity is empirically grounded; Kolmogorov [18] adopted σ-additivity as an axiom.

In quantum theory the same tension recurs in a different guise. The algebra of propositions about a quantum system is not Boolean: incompatible observables cannot be jointly determined, and the closed subspaces of a Hilbert space H form an *orthomodular lattice* $L(H)$ in which distributivity may fail. The classical extension machinery — Carathéodory's outer measure, which needs a Boolean algebra of sets — does not apply. For the single lattice $L(H)$ this is rescued by Gleason's rigidity theorem [11], but Hilbert space is special: for general OMLs no analogue is known.

This note asks the shared question — *when does finite coherence force countable behaviour?* — in the OML setting. Our organising observation is that the question, vague as usually posed, splits into two axes that the Boolean case fuses: an *extension* axis, classical and settled but for one residual case, and a *descent* axis that is genuinely open. The survey is built around that split.

The relational standpoint. We work in the spirit of the "structure from observation" programme [19]: rather than begin with a space Ω of outcomes and ask which σ-algebras it supports, begin with the propositions and their entailment order, and ask what that relational structure alone determines. For non-Boolean algebras the natural dual is not a space of points but the McDonald–Bimbó space of *filters* (§2.3). This is what makes the open problem worth its difficulty: a positive answer would be a *point-free* (relational) σ-additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory (§5).

2 Preliminaries

We collect the lattice-theoretic apparatus; all notions are standard.

2.1 Orthomodular lattices, and the orthogonal/meet-zero gap

Definition 2.1 (Orthocomplemented lattice)**.** A *bounded lattice* is a partially ordered set in which every pair a, b has a join $a \vee b$ and meet $a \wedge b$, with greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^\perp$ satisfying, for all a, b:

(i) $a \wedge a^\perp = \mathbf{0}$ and $a \vee a^\perp = \mathbf{1}$ (complement),
(ii) $a^{\perp\perp} = a$ (involution),
(iii) $a \leq b \Rightarrow b^\perp \leq a^\perp$ (order-reversing).

Definition 2.2 (Orthogonal; meet-zero)**.** Elements a, b are *orthogonal*, written $a \perp b$, if $a \leq b^\perp$ (equivalently $b \leq a^\perp$). They are *meet-zero* if $a \wedge b = \mathbf{0}$. Orthogonality always implies meet-zero; the converse holds in every Boolean algebra but **fails** once the lattice is non-distributive. This gap underlies the distinctions developed below.

Definition 2.3 (Orthomodular lattice)**.** An orthocomplemented lattice A is *orthomodular* if

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^\perp \wedge b). \tag{1}$$

A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is $L(H)$, the closed subspaces of a Hilbert space with $a^\perp$ the orthogonal complement.

Example 2.4 (MO_n, and the gap made concrete)**.** MO_n is the lattice with $\mathbf{0}$, $\mathbf{1}$, and n complementary pairs of incomparable atoms, all sharing only $\mathbf{0}$ and $\mathbf{1}$. It is orthomodular and non-distributive ($n \geq 2$): two atoms drawn from *distinct* complementary pairs have $a \wedge b = \mathbf{0}$ yet are *not* orthogonal, so meet-zero $\neq$ orthogonal already here. MO_3 is the smallest non-distributive OML.

Definition 2.5 (OMP; concrete logic)**.** An *orthomodular poset* (OMP) weakens Definition 2.3: joins $a \vee b$ are required only for *orthogonal* pairs. An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into $(\mathcal{P}(X), \subseteq, X \setminus (-))$ for some set X, preserving existing orthogonal joins.

Remark 2.6 (Concreteness does not close the gap). A set-representable OMP needs only closure under complement and *disjoint* union [2]; intersection-closure would force a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L-member inside $A \cap B$, so orthogonality ($A \cap B = \varnothing$) implies meet-zero, but not conversely. Indeed MO_3 is *itself* concrete (it carries an order-determining set of two-valued states; Gudder), so the paradigmatic meet-zero-$\neq$- orthogonal example is a concrete logic. The condition that closes the gap is not concreteness but a richness/atomicity condition (meet-zero coincides with orthogonality iff every nonempty $A \cap B$ contains a nonzero L-member); the abstract form $a \wedge b = \mathbf{0} \Rightarrow a \perp b$ is Tkadlec's *Boolean orthoposet* condition (which does *not* mean Boolean *algebra*).

2.2 The state/meet-additive/charge ladder

Definition 2.7 (State; meet-additive state; charge-extendible)**.** Let A be an OML. A *state* is a map $s : A \to [0, 1]$ with $s(\mathbf{1}) = 1$ that is additive on *orthogonal* pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$. Three strengthenings, strictly increasing in general (e.g. on MO_3, Example 2.8):

(1) *State* — additive on orthogonal pairs.
(2) *Meet-additive* — additive on every *meet-zero* pair $a \wedge b = \mathbf{0}$ (binary, strictly stronger). We avoid the word "valuation" here: in lattice theory it denotes the modular function $v(a) + v(b) = v(a \wedge b) + v(a \vee b)$, a different condition.
(3) *Charge-extendible* — the n-ary condition $\sum_i s(a_i) \le 1$ for every pairwise-meet-zero family $\{a_i\}$ (strictly stronger again). This is the *extension* condition of §3.

Example 2.8 (The ladder is strict)**.** On MO_3 with complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$, the maximally mixed $s \equiv \frac{1}{2}$ is meet-additive (clears (2)) yet fails (3) at $\{p, q, r\}$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So no state on MO_3 is charge-extendible.

2.3 The McDonald–Bimbó dual space

Definition 2.9 (Dual space; representation map)**.** The McDonald–Bimbó duality [20] assigns to an OML A a compact space

$$S_0(A) = \big(F(A), \subseteq, \perp_A, P(A), T\big),$$

where $F(A)$ is the set of *filters* of A, ordered by $\subseteq$ and carrying an orthogonality relation $\perp_A$ and a Stone topology T, and $P(A) \subseteq F(A)$ is the set of *principal* filters — the "physical points," the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For $a \in A$ set $h(a) = \{x \in F(A) : a \in x\}$; then h is an OML isomorphism onto the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^\dagger$. Define $\mu(h(a)) = s(a)$.

Remark 2.10 (The duality is finitary — the wall)*.* The identity $h(\bigvee_n a_n) = (\bigcup_n h(a_n))^{\perp\perp}$ holds for *finite* joins but fails for countable ones: finitary OML homomorphisms need not preserve infinite suprema. This finitariness is the wall the open problem of §5 runs into. (It is a *countable*-join phenomenon, not specific to joins: since $a \mapsto a^\perp$ is an order anti-isomorphism, $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$, so the failure to preserve countable meets is its De Morgan dual — the wall obstructs countable meets and joins together.) The rival OML duality of Cannon–Döring [3] is also finitary and carries no measure; we use McDonald–Bimbó because it carries $P(A)$ as explicit structure.

2.4 The Boolean engine: Stone duality

In the Boolean case, the descent argument ("σ-additive $\Leftrightarrow$ the measure concentrates on the physical points") runs on *Stone duality*: a Boolean event algebra embeds in the clopen algebra of its Stone space, a charge extends to a Baire measure there (Carathéodory), and σ-additivity is equivalent to concentration of that measure on the realised points [19]. Standing behind the σ-additive step is the *Loomis–Sikorski* representation — a σ-complete Boolean algebra is a σ-tribe of sets modulo a σ-ideal — which is what makes the countable-join bookkeeping go through. There is *no* OML analogue of either, and supplying one — or proving none exists — is the open problem of §5: the Boolean proof has an engine; the OML proof needs one and does not have it.

3 Two axes: extension and descent

The central question — *when does a state s on an OML A extend to a σ-additive measure on $S_0(A)$?* — splits into two **independent** axes that the Boolean case fuses. A state s on A faces:

	Extension axis	Descent axis
Question	Does μ extend to a finitely additive charge on the *full* Boolean clopen algebra of $S_0(A)$?	Once extended, does the measure concentrate on the physical points $P(A)$?
Governed by	the meet-zero/orthogonal gap (§2.1)	σ-additivity — needs the missing engine (§2.4)
Status	**settled**: Horn–Tarski / Pitowsky feasibility; finite case = decidable LP; can fail for every state.	**open**. *The descent residue.*

Two observations pin down the extension axis.

Proposition 3.1 (Finite case)**.** *For finite A the extension condition is a decidable linear program (Horn–Tarski [15] / Pitowsky correlation-polytope feasibility [22]); since h preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy $\sum_i s(a_i) \le 1$ over pairwise-meet-zero families. It can fail for* every *state (*MO_3*, Example 2.8).*

Proposition 3.2 (Infinite $L(H)$, normal states)**.** *Let $\dim H = \infty$. Any state s on $L(H)$ with $s(e) > 0$ for some finite-dimensional e fails to extend to a charge on $S_0(L(H))$; in particular no normal (Gleason) state extends.*

Proof sketch. Since $s(e) > 0$ on some finite-dimensional e, it is positive on a 2-plane inside it; fix such a 2-plane e_0, with $p^\perp$ the orthocomplement of a line p *within* e_0. Choose k lines $p_1, \dots, p_k \subset e_0$ no two of which are orthogonal; then the $2k$ lines $\{p_i, p_i^\perp\}$ are distinct and pairwise meet-zero (distinct lines meet at $\mathbf{0}$), so h sends them to pairwise-disjoint clopens. Orthoadditivity gives $s(p_i) + s(p_i^\perp) = s(e_0)$ for each i, and a charge on the disjoint clopens forces $\sum_i [s(p_i) + s(p_i^\perp)] = k\, s(e_0) \le 1$ for every k. If $s(e_0) > 0$, take $k > 1/s(e_0)$. The argument is finitary; σ-completeness of the lattice is irrelevant. □

Lift s to a bounded functional via Mackey–Gleason / Bunce–Wright [1] (valid as $B(H)$ has no type I_2 summand) and apply the Takesaki normal/singular decomposition $s = s_{\mathrm{n}} + s_{\mathrm{sing}}$. Any state with $s_{\mathrm{n}} \neq 0$ has $s(e) > 0$ on some finite-dimensional e and is killed by Proposition 3.2; the only survivors are the *purely singular* states ($s_{\mathrm{n}} = 0$, equivalently $s(e) = 0$ for all finite-dimensional e; e.g. ultrafilter vector states, or pullbacks of Calkin-algebra states). So on the extension axis the sole survivors are the purely singular states of $L(H)$, treated in §6. The genuine open problem lives on the descent axis (§5).

4 What is known

The extension axis is settled apart from the singular case of §6; the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. We survey what the literature supplies on that question — the Boolean benchmark it would have to match, the partial OML results, and the routes already known to be blocked.

4.1 The Boolean benchmark

In the Boolean case the Kelley–Vladimirov–Pták characterisation (Fremlin [9], Thm 391D) is complete: a Boolean algebra carries a strictly positive σ-additive measure iff it is Dedekind σ-complete, weakly (σ, ∞)-distributive, and chargeable. Each condition has *no* clean OML analogue: weak (σ, ∞)-distributivity relies on distributivity of $\bigwedge$ over $\bigvee$; chargeability has no structural characterisation on OMLs.

4.2 Positive and obstructive OML results

The positive results require structure rich enough to approximate Hilbert-space geometry: Gleason [11] ($L(H)$, dim ≥ 3; σ-additivity *on the lattice*), Bunce–Wright [1] (JBW projection lattices), Chetcuti–Dvurečenskij [4]. The obstructive results show the Boolean machinery cannot be transplanted: Pták–Pulmannová [24] (conditions forcing σ-additivity collapse the lattice to Boolean), Navara [21] (regularity does not force σ-additivity), Pták [23] (exotic state spaces via Greechie pasting).

4.3 Three blocked routes to a σ-OML engine

The descent argument needs an OML analogue of the Boolean engine — a σ-Stone duality, or equivalently a Loomis–Sikorski representation. The three known routes are all blocked.

(i) **RDP / effect-algebra.** Loomis–Sikorski holds for σ-MV algebras [7] and monotone σ-effect algebras *with* the Riesz Decomposition Property [8]; but a lattice effect algebra has RDP iff it is MV, and OML $\cap$ MV = Boolean (Kalmbach [16]). So OML+RDP $\Leftrightarrow$ Boolean.

(ii) **MacNeille completion.** The MacNeille completion of an OML need not be orthomodular (Harding [12]); one cannot complete to absorb countable joins.

(iii) **σ-Stone duality.** None exists. McDonald–Bimbó is finitary (Remark 2.10); Freytes' theory [10] of σ-complete OMLs is equational, with no set/tribe representation.

4.4 Prior art on point-free quantum probability

Every existing point-free or directed-context construction lands on one side of the σ-additivity-on-a-non-distributive-structure line.

Gleason tradition
Varadarajan, Kalmbach, Pták–Pulmannová: σ-additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

Döring 2009 [6] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

Bohrification (HLS)
[13, 14] point-free and the internal valuation *is* σ-additive (Scott-continuous) — but factors through an internally *distributive* locale, σ-additive only per commutative context. Non-distributivity is tamed, not carried natively.

Localic valuations
Vickers, Simpson, Coquand–Spitters: point-free σ-additive, but on *frames* — distributive by definition.

OML Stone dualities
McDonald–Bimbó [20] and Cannon–Döring [3]: purely structural, no measure, no σ-version.

The unoccupied conjunction — point-free $\times$ σ-additive $\times$ natively non-distributive $\times$ directed-system-built — is exactly the descent residue. (The directed-context-yields-non-distributivity *idea* is itself not new; it is the shape of Bohrification and of colimit-over-Boolean-contexts generation of OMLs. Novelty would lie in the σ-additive-point-free-native combination.)

Remark 4.1 (The boundary of the negative results). Two qualifications sharpen what is and is not ruled out. *(i) No impossibility theorem* covers the *concrete* class: the known no-go results are finite or two-valued, and the one positive occupant $L(H)$+Gleason is *non-concrete* (Kochen–Specker [17], dim ≥ 3: $L(H)$ admits no two-valued states at all, a fortiori none order-determining), so concreteness is the qualifier on which the open frontier turns. *(ii) The absence of a σ-Loomis–Sikorski* is visible in the dualities themselves: both take infinite joins as a *closure*, not a union (Cannon–Döring $\mathrm{cls}(\bigcup S_i)$; McDonald–Bimbó $(\bigcup h(a_n))^{\perp\perp}$). One should not overstate this: Gudder concrete logics *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a σ-Loomis–Sikorski theorem.

5 The open problem

Open Problem. Does a σ-complete, concrete, non-Boolean, infinite OML admit a Loomis–Sikorski-type representation (a σ-tribe of sets modulo a σ-ideal), or a countable-join-preserving σ-Stone duality — or can one prove that none exists?

A positive answer is the engine for *relational probability without realisations*: σ-additive probability built from the entailment relation of a non-distributive OML, not descended from any sample space — the non-Boolean analogue of localic (pointless) measure theory. A negative answer (an impossibility theorem) closes the whole cluster, including both residues, at once. Either way the outcome is decisive: no impossibility theorem is known for the live class (Remark 4.1), and all three constructive routes are blocked (§4.3), so the state of the art is not "needs a new idea" but "needs a representation bypassing all three blocked routes."

Remark 5.1 (Why $L(H)$+Gleason does *not* already settle this — the (A)/(B) point-space equivocation). The motivating idea ("probability from relations, point-free") may *look* delivered by $L(H)$ + Gleason. It is not, and the reason is a genuine equivocation between two point-spaces. **(A)** the Hilbert rays of H, from which the Gleason density ρ in $s = \mathrm{tr}(\rho\,\cdot)$ is built; **(B)** the McDonald–Bimbó dual filters $P(A)$. The programme defines *realisation* as concentration on **(B)**. Gleason removes **(B)** but *requires* **(A)** — and is a *representation* theorem, reducing every lattice state to the point datum ρ, the most point-ful result available. So $L(H)$/Gleason exhibits a *separation* — a σ-additive measure exists on the lattice, while (by Proposition 3.2) no normal state even extends to a charge on $S_0(L(H))$, so a fortiori none concentrates on the dual points $P(A)$ — not point-freeness, and it cannot serve as the existence proof sought. Because it cannot, the σ-OML representation above is the *only* candidate engine; it would be an error to regard Gleason's theorem as already furnishing a relational, point-free probability.

6 Directions

Two lines of attack present themselves, both bounded by the same finitary-to-σ obstruction.

The general construction

Construct a Loomis–Sikorski-type representation, or a countable-join-preserving σ-Stone duality, for a concrete non-Boolean infinite σ-OML — or prove that none exists. This resolves the open problem directly, but no partial construction is currently in hand.

The singular case for $L(H)$

Determine whether a *singular* orthoadditive state on $L(H)$ extends to a charge on $S_0(L(H))$ — a question about one specific lattice, accessible to operator-algebraic methods (Calkin, Bunce–Wright, Takesaki), and conditionally self-contained (subject to the finite-versus-countable question below).

The singular case is the more tractable entry point, and it is informative whichever way it resolves. If a singular state fails to extend, then by the dichotomy following Proposition 3.2 *no* state on $L(H)$ extends; since the singular states are the only remaining candidates for a state whose descent behaviour is independent of extension, this would settle that subsidiary question negatively. If a singular state does extend, it is the first explicit example of a state that extends but may fail to concentrate on $P(A)$ — the motivating example the general construction lacks.

Whether the singular case is genuinely more tractable, however, turns on a prior question.

Open Problem (Finite versus countable). Is the obstruction to extending a singular state on $L(H)$ realised by a *finite* or only by a *countable* pairwise-meet-zero family?

Proposition 3.2 is finitary — it uses finitely many lines in a single 2-plane — which is why it avoids the descent-axis obstruction entirely. For singular states the relevant pairwise-meet-zero families are infinite-dimensional subspaces with trivial intersection, and the line argument is vacuous there ($s(e) = 0$ on every finite-dimensional e). That the line argument is vacuous does not, however, imply that the obstruction requires a countable family: whether a *finite* family of infinite-dimensional, pairwise-meet-zero subspaces can force $\sum_i s(a_i) > 1$ is unsettled. If such a finite obstruction exists, the singular case is a self-contained operator-algebraic problem, separable from the general construction; if only a countable family obstructs, it is governed by the same finitary-to-σ passage as the general case. The question therefore reduces, concretely, to whether orthoadditivity of a singular state forces $\sum_i s(a_i) > 1$ on some finite family of infinite-dimensional, pairwise-trivially-intersecting subspaces of H.

References

[1] Leslie J. Bunce and J. D. Maitland Wright. The Mackey–Gleason problem. *Bulletin of the American Mathematical Society*, 26(2):288–293, 1992.

[2] Dominika Burešová and Pavel Pták. On the set-representable orthomodular posets that are point-distinguishing. *International Journal of Theoretical Physics*, 62, 2023. Also arXiv:2401.13798.

[3] Sarah Cannon. *The Spectral Presheaf of an Orthomodular Lattice: A Generalisation of Stone Duality to Orthomodular Lattices.* PhD thesis, University of Oxford, 2013. See also Cannon and Döring, "A generalisation of Stone duality to orthomodular lattices," in *Contemporary Logic and Computing* (2018).

[4] Emmanuel Chetcuti and Anatolij Dvurečenskij. On the Gleason theorem for effect algebras. *Mathematica Slovaca*, 55(3):297–308, 2005.

[5] Bruno de Finetti. *Theory of Probability*, volume 1. Wiley, 1974.

[6] Andreas Döring. Quantum states and measures on the spectral presheaf. *Advanced Science Letters*, 2(3):291–301, 2009. arXiv:0809.4847.

[7] Anatolij Dvurečenskij. Loomis–sikorski theorem for σ-complete MV-algebras and ℓ-groups. *Journal of the Australian Mathematical Society*, 68(2):261–277, 2000.

[8] Anatolij Dvurečenskij. Loomis–sikorski theorem for monotone σ-complete effect algebras. *Journal of the Australian Mathematical Society*, 79(3):305–318, 2005.

[9] D. H. Fremlin. *Measure Theory, Vol. 3: Measure Algebras.* Torres Fremlin, Colchester, 2004.

[10] Hector Freytes. An equational theory for σ-complete orthomodular lattices. *Soft Computing*, 24(14):10257–10264, 2020.

[11] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.

[12] John Harding. Orthomodular lattices whose MacNeille completions are not orthomodular. *Order*, 8(1):93–103, 1991.

[13] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. A topos for algebraic quantum theory. *Communications in Mathematical Physics*, 291(1):63–110, 2009. arXiv:0709.4364.

[14] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. Bohrification. In Hans Halvorson, editor, *Deep Beauty: Understanding the Quantum World through Mathematical Innovation*, pages 271–313. Cambridge University Press, 2011. arXiv:0909.3468.

[15] Alfred Horn and Alfred Tarski. Measures in Boolean algebras. *Transactions of the American Mathematical Society*, 64(3):467–497, 1948.

[16] Gudrun Kalmbach. *Orthomodular Lattices*. Academic Press, 1983.

[17] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.

[18] A. N. Kolmogorov. *Foundations of the Theory of Probability*. Chelsea, 1933.

[19] Patrick S. Mahon. When does observational coherence determine probability? Preprint, 2026.

[20] Joseph McDonald and Katalin Bimbó. A Stone-type duality for orthomodular lattices. *Mathematical Logic Quarterly*, 2023. arXiv:2208.07430.

[21] Mirko Navara. When is the integral on quantum probability spaces additive? *Bulletin of the Polish Academy of Sciences — Mathematics*, 40(1):29–33, 1992.

[22] Itamar Pitowsky. *Quantum Probability — Quantum Logic*, volume 321 of *Lecture Notes in Physics*. Springer, Berlin, 1989.

[23] Pavel Pták. Exotic logics. *Colloquium Mathematicum*, 54(1):1–7, 1987.

[24] Pavel Pták and Sylvia Pulmannová. *Orthomodular Structures as Quantum Logics*, volume 44 of *Fundamental Theories of Physics*. Kluwer, Dordrecht, 1991.